

Manual

Material Buffers and Inventories MPL-MPB 8.1

Version 1.0.54

Last changed on: 19.06.2020

Copyright

©Copyright 2012 All rights reserved.

SAP® and R/3® are registered trademarks of SAP AG.

WINDOWS® is a registered trademark of Microsoft Corporation.

MPDV® and HYDRA® are registered trademarks of MPDV Mikrolab GmbH.

ORACLE® is a registered trademark of ORACLE Corporation, California, USA.

Copying and distribution of this documentation or any part thereof, for any purpose or in any form, is prohibited without prior written permission from MPDV Mikrolab GmbH.

The information contained in this documentation is subject to change without prior notice.

Contents

1	Overview of Material Buffers and Inventories	4
2	Material Buffers	6
3	Transport Units.....	10
4	1 Assignment of Transport Unit to Material Type.....	12
5	Inventory Limits	14
6	Stock Monitoring	16

1 Overview of Material Buffers and Inventories

Possible fields of application

The material buffer and inventories function package allows for material stock to be managed and evaluated.

Implementation notes

The function package "material buffers and inventories" is used if you

- use the function packages "material and inventory management", "batch data management" or "batch processing" and
 - would like to enhance them by managing storage locations.
 - would like to record which anonymous, logistical containers include or transport the material.
- would like the system to calculate the consumption of posted input materials (retrograde).
- would like to enter the consumption of posted input materials explicitly.

Integration

- Input and output material buffers can be defined at the object "machine/workplace".

Functions

- Configuration of material buffers
 - Functions to configure material buffers
- Configuration of transport units
 - Editing function for transport units to classify the means of transportation, such as skeleton containers, pallets, containers and to manage stock quantities.
 - Editing function to classify material types in transport units
- Configuration of inventory limits
 - Editing function for inventory limits to manage minimum and maximum stock levels for the material included in the material buffer
- Management of buffer inventories
 - Management of material stock relating to material buffers or transport units
- Assignment of material buffer to the batch
 - Appropriate collection and posting functions
- Posting to input batches
 - Retrograde or discrete withdrawal of consumption from registered input batches subject to the configuration

- Repost
 - Reposting of individual batches

2 Material Buffers

Menu	Master data → Material → Material buffers
Transaction code	mbuf
Function authorization	mbuf

Usage

This function is used to create or modify material buffers in the system.

Integration

Material buffers are used in the system to represent both actual physical and purely virtual material storage or areas in which materials are kept for temporary storage in the production process. Material buffers are identified by unique keys.

Selection criteria

The following selection criteria are available in the application:

Material buffer

Selection using the [material buffer](#) Ident

Designation

Selection using the designation of the [material buffer](#)

Type

Selection using the material buffer type. Permissible values are:

- F – Production buffer
- H – Hierarchical buffer
- W - Goods receipt buffer
- C – casting buffers in terms of composition

Cost center

Selection using cost centers to which the [material buffers](#) were assigned.

Company

Selection using companies to which the [material buffers](#) were assigned.

Department

Selection using departments to which the [material buffers](#) were assigned.

Area

Selection using areas to which the [material buffers](#) were assigned.



When using multiple selection criteria - if nothing else is specified - the amount of overlap of the selection criteria is displayed.

Field descriptions – general

Material buffer

Ident of the [material buffer](#) to be created or edited.

Type

Material buffer type. Permissible values are:

- F – Production buffer
Production buffers are material buffers that are located within the production areas. Storage locations or material buffers for resources should always be configured as production buffers.
- H – Hierarchical buffer
A hierarchical buffer combines several individual buffers. In the evaluations, the cumulative quantity of the individual buffers is represented in the hierarchical buffer.
- W – Goods receipt buffer
The material buffers in which material from external sources is delivered are classified as goods receipt buffers. They are the source for the material consumption in production.

On the processing side, buffers of the production and goods receipt types behave in the same way. In contrast, a hierarchical buffer is a 'virtual' buffer that enables one or more buffers (hierarchical buffers as well) to be compiled in order evaluate them in in this way using the function [stock overview](#) (menu *Material management > Inventory management > Stock overview*).

Designation

Long text regarding [material buffer](#)

Storage location

Reference to the physical storage location of the material buffer (no current use on the processing side)

Department

Department to which the [material buffer](#) belongs

Area

Area to which the [material buffer](#) belongs

Cost center

Cost center to which the [material buffer](#) belongs

Company

Company in which the [material buffer](#) is located

Comment

Additional text

Inventory management tab

Recycle bin

All material buffers defined as recycle bins are cyclically deleted or archived depending on the configuration or customizing. The process that deletes or archives batches in the material buffers is integrated into the scheduler. The period after which the batches are deleted or archived is configured using data management.

Batches that are located in a material buffer identified as a "recycle bin" can no longer be logged on. In this case, they must be manually reposted in another material buffer.

Batches cannot be logged on as input batches at machine with an incoming material buffer, which is configured as recycle bin.

Include in stock

Definition regarding whether the batches located in this material buffer are to be included in the stock overview or not.

Retention period

Duration of the storage period in the buffer (no current use on the processing side).

Hierarchy tab

Hierarchy

Indicates the hierarchy level assigned to the buffer.

Hierarchical buffer

Indicates the assignment to the superior hierarchical buffer. Note that a buffer can only be assigned to a hierarchical buffer with a hierarchy (number) that is greater than that of the assigned buffer.

Batch transport tab

Type

- No buffer

This has to do with one of the buffer versions described in the following.

- Input buffer, output buffer

In order to configure the batch transport across facilities, a specification can be made here indicating if the material buffer is an input or output buffer. A corresponding system is defined for an output buffer with which specifications are made regarding where that data posted in this material buffer are transported.



If you configure a material buffer as “input buffer“, the transport status of the batches posted to this buffer will be set to “I“ = initial. Consequently, these batches cannot be logged on to AIP.

Virtual stock buffer

This field can be used if a material buffer is defined as a stock posting buffer. In this way, an additional specification can be made regarding whether the material buffer is a virtual stock buffer or not.

This identification of the material buffer as a placeholder for an external inventory system is only necessary in order to make it possible to query the inventory for the virtual stock at a later time. This identification does not have any current significance.

Corresponding system

By defining the corresponding system, it is specified where all batches that are included in this material buffer are to be transferred. In the case of transport output buffers and stock posting buffer, the corresponding system is defined here; a specification is made regarding the system to which the batches are transferred.

Stock posting buffer In order to carry out stock postings, a material buffer can be defined as a stock posting buffer. The transfer is made using the standard batch interface with a file that is created in the file system in a directory specified by the corresponding system.

3 Transport Units

Summary

Menu	Material → Master data → Transport units
Transaction code	tu
Function authorization	tu

Usage

Use this function to create or to change transport units in the system.

Transport units are defined as those load carriers that are used to transport materials. When output batches are recorded at the terminal, the assignment to the producing [material type](#) can serve as selection that is saved to the batches.

Integration

The transport units will be shown in the evaluations in material management.

Selection criteria

The following selection criteria are available in the application:

Transport unit

Only the selected transport unit will be used.

Field descriptions

Transport unit

Unique identification of a transport unit

Designation

Clear text description of the transport unit

Quantity

Quantity of available transport units

Width

Width of the transport unit in the assigned unit. Currently used only for terminal display features.

Height

Height of the transport unit in the assigned unit. Currently used only for terminal display features.

Length

Length of the transport unit in the assigned unit. Currently used only for terminal display features.

Weight

Weight of the transport unit in the assigned unit. Currently used only for terminal display features.

Inventory management

Indicator relating to the inventory management of the transport unit. If this indicator is not set, this transport unit will also not be taken into account in the evaluation of the stock overview transport units

(Menu: *Material management* > *Inventory management* > *TPU stock overview*).

4 1 Assignment of Transport Unit to Material Type

Summary

Menu	Master data → Material → TPE assignment - Material type
Transaction code	astrum
Function authorization	astrum

Usage

This function is used to assign the corresponding [transport units](#) to a [material type](#) in the system.

Integration

For each [material type](#), several possible [transport units](#) can be defined. Then, when forming an output batch, the corresponding [transport unit](#) can be selected from those for the [material type](#) of the material to be produced.

Requirement

The transport units must be defined in the system.

Editing functions

Copy

There is the functionality to copy assignments from one material type to another. It can be chosen to either copy all assignments or only the missing ones.

Selection criteria

The following selection criteria are available in the application:

Material type

Only materials with the selected [material type](#) are selected.

Transport unit

Only materials with the selected [transport unit](#) are selected.

When using multiple selection criteria - if nothing else is specified - the amount of overlap of the selection criteria is displayed.

Field descriptions

Material type

Designation of the assigned [transport unit](#).

Transport unit

Designation of the assigned [transport unit](#).

Standard

The preferred [transport unit](#) for transporting materials of the specified material type.

Quantity

Quantity of the materials of this [material type](#) that match with this [transport unit](#). Used in the evaluation overview of [transport units](#) available.

Unit

Not in current use.

5 Inventory Limits

Summary

Menu	Master data → Material → Inventory limits
Transaction code	invl
Function authorization	invl

This function is used to create or modify inventory limits in the system.

Usage

This dialog is used to assign those materials to the [material buffers](#) that are temporarily stored in this [material buffer](#). The assignment enables the minimum stock to be specified in the buffer.

Integration

The values stored here are used in the following functions:

- Material management → [Stock monitoring](#)
- Operating facilities management → Graphic [machinery](#)
- Production control → [Graphic planning](#)

Requirement

[Material buffer](#) and units must be defined.

Selection criteria

The following selection criteria are available in the application:

Material buffer

Displays all inventory limits for the selected material buffer.

Material

Displays all inventory limits for the selected material.

When using multiple selection criteria, the amount of overlap of the selection criteria is displayed.

Field descriptions

Material

Unique material key

MBuffer

Name of the assigned material buffer

Unit

Quantity unit in which the inventory is listed

Minimum stock

Minimum stock of the material in the material buffer

Alert Min. stock level

Quantity of stock at which an alert occurs for increasing inventory

Min. warn. stock level

Quantity of stock at which a warning occurs for checking inventory

Max. warn. stock level

Quantity of stock at which a warning occurs for checking inventory

Alert Max. stock level

Quantity of stock at which an alert occurs for decreasing inventory

Max. stock level

Maximum material stock level in buffer

Comment

Additional comments

6 Stock Monitoring

Summary

Menu	Material management → Inventory management → Stock monitoring
Transaction code	stomo
Function authorization	stomo

Usage

Stock monitoring allows to display those materials (batches), for which the [inventory limits](#) are exceeded or not reached in a [material buffer](#).

Integration

Since these material buffers can have different sizes and since the definition of when such a material buffer is full depends on the size of the material, the definition of the inventory limits to be monitored is referred to a material and to a [material buffer](#). This creates the possibility to monitor only critical materials and/or only certain material buffers.

All materials per material buffer that match the entered selection criteria and for which [inventory limits](#) were defined for the respective material buffer will be shown in this list that updates cyclically. To do so, the existing batches will be cumulated to the material number and be displayed. To further limit this, only batches of a specific status can be used.

Requirements

The following requirements must be met to display materials in stock monitoring:

1. In the configuration (menu: *Master data > Material > Inventory limits*) you must have defined corresponding inventory limits for the [combination of material and material buffer](#).
2. The [material buffer](#) (menu: *Master data > Material > Material buffer*) must not be a hierarchical buffer. In addition the indicator "include in stock" must be set.

Selection criteria

The following selection criteria are available in the application:

Material buffer

This selection criterion refers to the material buffer of the material. All materials with the selected [material buffer](#) will be shown.

Material

Any stock with the selected material will be shown.

Batch status

This selection criterion refers to the batch status of the stocks. All materials with the selected batch status will be shown.

If several selection criteria are used, the intersection of the selection criteria will be shown.

Field descriptions

Fill level

Symbol	Symbol color	Meaning	Meaning related to the configured inventory limits
	green	Stock o.k.	Stock between min. warning stock level and max. warning stock level.
	yellow	Stock below warning limit	Stock between min. warning stock level and minimum stock.
	red	Stock below alarm limit	Stock between min. alarm quantity and min. warning stock level
	red	Stock below minimum	Stock below minimum stock
	yellow	Stock above warning limit	Stock between max. warning stock level and max. alarm quantity.
	red	Stock above alarm limit	Stock between max. alarm quantity and max. stock level
	red	Stock above maximum	Stock above max. stock level

Material type

Assigned material type of the material

Material

Material number

Material designation

Designation of the material

Quantity

Stock quantity of the material

Unit

Quantity unit

Material buffer

Assigned material buffer of the material

Designation

Designation of the material buffer

Company

Assigned company to the material buffer

Department

Assigned department to the material buffer

Area

Assigned area to the material buffer

Cost center

Assigned cost center to the material buffer

Storage location

Assigned storage location to the material buffer

Type

Type of the material buffer

Min. stock quantity

Defined min. stock quantity of the material in this material buffer

Lower alarm limit (min. alarm quantity)

Defined min. alarm quantity of the material in this material buffer

Lower warning limit (min. warning stock level)

Defined lower warning stock level of the material in this material buffer

Upper warning limit (max. warning stock level)

Defined upper warning stock level of the material in this material buffer

Upper alarm limit (max. alarm quantity)

Defined upper alarm quantity of the material in this material buffer

Max. stock level

Defined max. stock level of the material in this material buffer